Mayur Rishi

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Education

Army Institute of Technology | *Bachelor of Engineering — Electronics & Telecommunication* *Pune, Maharashtra · June 2026*

CGPA: **8.43 / 10** | Q2 International Peer-Reviewed Publication, ETASR (Switzerland) · Patent Filing in Progress

Skills

Programming	Python, TypeScript, JavaScript, SQL
AI & LLM	LangChain, LangGraph, CrewAI, RAG, Agentic AI, Prompt Engineering, MCP
Backend	FastAPI, Django, REST APIs, Pydantic, Alembic, background job queues, Apache Kafka
Frontend	React, Next.js (App Router), Tailwind CSS
Vector DBs & ML	ChromaDB, FAISS, SentenceTransformers, Scikit-Learn
Databases & Storage	PostgreSQL, MySQL, SQLite, Redis, SQLAlchemy, Cloudflare R2 (S3-compatible)
Cloud & DevOps	Docker Compose, VPS deployment, Razorpay integration, AWS
Evaluation & Testing	LangSmith, Ragas, Pytest (including access-control tests)
Tools	Git, JIRA, Confluence
Methodologies	Agile, Scrum

Freelance Services

- **Full-stack web apps:** Next.js with FastAPI or Django and PostgreSQL — authentication, role-based dashboards, admin consoles, Razorpay payments and secure large-file uploads (R2/S3).
- **AI & LLM solutions:** RAG chatbots, multi-agent workflows (LangGraph, LangChain, CrewAI) and LLM evaluation and tracing with Ragas and LangSmith.
- **Backend & integrations:** REST APIs, background job queues, PDF and certificate generation, audit logging and third-party API integrations.
- **Deployment & handover:** Docker Compose on a VPS, database backups and staged rollouts, with tests and runnable deployment steps delivered with each feature.
- **How I work:** plain-language request → plan → working feature, with trade-offs (cost, security, effort) explained before building.

Work Experience

Forward Deployment Engineer Trainee | *Prodapt Solutions* *Jun 2026 – Present*

- Completed enterprise-grade Forward Deployment Engineering training modules in applied AI and modern backend systems.
- Built and deployed production-style AI projects using Agentic AI, LangGraph, LangChain, and FastAPI.
- Developed RAG and multi-agent backends with REST APIs, SQLAlchemy, SQLite, and modular architecture.

QA Engineer Intern | *Solytics Partners* *Remote · May 2025 – Nov 2025*

- Audited multi-source enterprise datasets, traced root causes, and reported findings that informed product decisions.
- Converted stakeholder requirements into actionable audit and test plans aligned with observed data.
- Collaborated with engineering, data, and product teams across Agile sprints.

App Developer Intern | *Front Benchers Infotech Limited* *Remote · Mar 2024 – Sept 2024*

- Translated client requirements for three product builds into plans and updates for non-technical stakeholders.
- Managed concurrent sprint deliverables from initial brief through production release.

Research & Projects

Spyrexo LMS — Learning & Internship Platform *Next.js · FastAPI · PostgreSQL · Cloudflare R2 · Docker*

- Built end to end and deployed on a VPS with Docker Compose: courses, live classes, mentor-graded internships, certificates and Razorpay payments, with student, teacher and admin roles (plus view-only admin) enforced centrally.
- Built an AI video-dubbing pipeline (Gemini Live Translate, Demucs, ffmpeg) that converts a video into Hindi, Telugu, Tamil and Kannada, run by a job queue with atomic claiming, retries and crash recovery.
- Implemented direct-to-R2 multipart uploads for multi-GB files with progress and signed download URLs, plus reference tracking so shared files are never deleted by mistake.
- Designed the internship documents workflow (offer and experience letters, certificates with reference numbers and verification tokens), audit logging and a payment ledger with CSV export; covered by Pytest, including access-control tests.

Aegis — AI Claims Intelligence Assistant *LangGraph · FastAPI · ChromaDB · Scikit-Learn · Kafka · React*

- Built a stateful 6-agent LangGraph workflow (retrieval, risk analysis, anomaly detection, summarization, investigation, advisor) with parallel fan-out and automatic handoff of high-risk claims to a forensic-investigation agent.
- Combined ChromaDB vector search, Random Forest risk scoring and Isolation Forest anomaly detection with real-time Apache Kafka ingestion and an adjuster AI copilot chatbot, grounded in the COIL 2000 insurance benchmark.
- Enforced human-in-the-loop governance: AI never approves payouts on its own; low-risk claims are auto-cleared and high-risk claims go to an adjuster for audited final sign-off.
- Scored 100% Hit@1/3/5 on retrieval, 0.9467 ROC-AUC (87.5% accuracy) on risk prediction and 100% factual grounding on its automated 5-criteria evaluation suite, at 13.5 s average end-to-end latency.

IoT-Based VR Haptic Glove — Independent Research *ETASR Q2 Publication · Patent Filing in Progress*

- Defined the methodology, cleaned sensor data, and published findings in a Q2 peer-reviewed international journal.
- Authored and defended the methodology, results, limitations, and implications for a technical audience.
- Progressing a patent filing for the research's novel sensor-fusion architecture.

NomNom Chat — Anonymous Campus Platform *Django, PostgreSQL, Redis*

- Analyzed post-launch behavior, iterated from feedback, and validated design assumptions against usage.
- Owned delivery from problem framing through deployment using Django, PostgreSQL, and Redis.